

PL/SQL Exercises

Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

Sql Command:

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE ApplySeniorCitizenDiscount()
-> BEGIN
->   DECLARE done INT DEFAULT FALSE;
->   DECLARE customer_id INT;
->   DECLARE customer_age INT;
->   DECLARE customer_dob DATE;
->
->   DECLARE cust_cursor CURSOR FOR
->     SELECT CustomerID, DOB
->     FROM Customers;
->
->   DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
->
->   OPEN cust_cursor;
->
->   read_loop: LOOP
->
->     FETCH cust_cursor INTO customer_id, customer_dob;
->
->     IF done THEN
->       LEAVE read_loop;
->     END IF;
->
->     SET customer_age = TIMESTAMPDIFF(YEAR, customer_dob, CURDATE());
->
->     IF customer_age > 60 THEN
->       UPDATE Loans
->       SET InterestRate = InterestRate - 1
->       WHERE CustomerID = customer_id;
->     END IF;
->
->   END LOOP;
->
->   CLOSE cust_cursor;
->
```

-> END\$\$

Query OK, 0 rows affected (0.01 sec)

mysql>

mysql> DELIMITER ;

mysql> SELECT CustomerID,

-> Name,

-> DOB,

-> TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age

-> FROM Customers;

```
+-----+-----+-----+-----+
| CustomerID | Name      | DOB      | Age |
+-----+-----+-----+-----+
| 1 | John Doe | 1985-05-15 | 41 |
| 2 | Jane Smith | 1990-07-20 | 35 |
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

NOTE:

-the procedure worked correctly because **no customer is older than 60**, so no update occurred.

Scenario 2: A customer can be promoted to VIP status based on their balance.

- **Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

SQL Command:

mysql> ALTER TABLE Customers

-> ADD IsVIP BOOLEAN DEFAULT FALSE;

Query OK, 0 rows affected (0.05 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> DELIMITER \$\$

mysql>

mysql> CREATE PROCEDURE UpdateVIPStatus()

-> BEGIN

-> DECLARE done INT DEFAULT FALSE;

-> DECLARE customer_id INT;

-> DECLARE customer_balance DECIMAL(10,2);

->

-> DECLARE cust_cursor CURSOR FOR

-> SELECT CustomerID, Balance

-> FROM Customers;

->

-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

->

-> OPEN cust_cursor;

->

-> vip_loop: LOOP

```

->
->     FETCH cust_cursor INTO customer_id, customer_balance;
->
->     IF done THEN
->         LEAVE vip_loop;
->     END IF;
->
->     IF customer_balance > 10000 THEN
->         UPDATE Customers
->         SET IsVIP = TRUE
->         WHERE CustomerID = customer_id;
->     END IF;
->
-> END LOOP;
->
-> CLOSE cust_cursor;
->
-> END$$

```

Query OK, 0 rows affected (0.01 sec)

```

mysql>
mysql> DELIMITER ;
mysql> SELECT CustomerID, Name, Balance, IsVIP
-> FROM Customers;

```

```

+-----+-----+-----+-----+
| CustomerID | Name      | Balance | IsVIP |
+-----+-----+-----+-----+
|      1 | John Doe | 1000.00 |    0 |
|      2 | Jane Smith | 1500.00 |    0 |
+-----+-----+-----+-----+

```

2 rows in set (0.00 sec)

NOTE:

-No one is VIP because both balances are below 10000.

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

SQL Command:

```

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE LoanReminder()
-> BEGIN
->     DECLARE done INT DEFAULT FALSE;
->     DECLARE customer_name VARCHAR(100);
->     DECLARE loan_due_date DATE;

```

```

->
-> DECLARE loan_cursor CURSOR FOR
->   SELECT c.Name, l.EndDate
->   FROM Customers c
->   JOIN Loans l
->   ON c.CustomerID = l.CustomerID
->   WHERE l.EndDate BETWEEN CURDATE()
->   AND DATE_ADD(CURDATE(), INTERVAL 30 DAY);
->
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
->
-> OPEN loan_cursor;
->
-> reminder_loop: LOOP
->
->   FETCH loan_cursor INTO customer_name, loan_due_date;
->
->   IF done THEN
->     LEAVE reminder_loop;
->   END IF;
->
->   SELECT CONCAT(
->     'Reminder: Dear ',
->     customer_name,
->     ', your loan is due on ',
->     loan_due_date
->   ) AS ReminderMessage;
->
-> END LOOP;
->
-> CLOSE loan_cursor;
->
-> END$$

```

Query OK, 0 rows affected (0.01 sec)

```

mysql>
mysql> DELIMITER ;
mysql> SELECT c.Name, l.EndDate
-> FROM Customers c
-> JOIN Loans l
-> ON c.CustomerID = l.CustomerID
-> WHERE l.EndDate BETWEEN CURDATE()
-> AND DATE_ADD(CURDATE(), INTERVAL 30 DAY);
Empty set (0.01 sec)

```

NOTE:

-which is not within the next 30 days, so the query will return 0 rows.